Universitat Pompeu Fabra Barcelona

Topic 5. Modulation and demodulation

In this topic

Topic 4 showed that a transmitter is a set of coordinates on an orthonormal basis of waveforms, and that the receiver recovers the noisy coordinates by correlation. It did not say which waveforms to use, nor how fast the coordinates can follow one another. This topic builds the waveforms of real systems. A pulse and its shifts carry a stream of symbols, a cosine and a sine carry two coordinates on a carrier, and a filter followed by a sampler replaces the bank of correlators. The price of each choice is paid in time and in frequency, and the Fourier transform measures it. Sections 5.1 to 5.4 correspond to the first lecture and Sections 5.5 and 5.6 to the second.

- We map bits to the symbols of a constellation, and we relate the symbol rate to the bit rate, the energy per symbol to the energy per bit, and both to the transmitted power.
- We build pulse amplitude modulation from a pulse whose shifts are orthonormal, and we recover the real Gaussian channel of Topic 4, one use per symbol.
- We compute every correlation with one matched filter and a sampler, and we see what intersymbol interference is when the sampling instants are wrong.
- We place two coordinates on a cosine and a sine at a carrier frequency, obtain PSK and QAM, and recover the complex Gaussian channel of Topic 1 with noise variance N_0.
- We define the bandwidth of a signal with the Fourier transform, and we state the Nyquist criterion: a pulse of period T needs a baseband bandwidth of at least $1/(2T)$.
- We measure the use of the spectrum by the spectral efficiency, and we place energy, bandwidth and error probability in one trade-off.

5.1 Symbols, rates and energies

A transmitter receives bits at a **bit rate** R_b, in bits per second, and sends symbols at a **symbol rate** $R_\mathrm{s} = 1/T$, in symbols per second, where T is the **symbol period**. The **mapper** takes the bits in groups of $\log_2 M$ and assigns to each group a point of a constellation $\mathcal{X}$ of M points, with a fixed **labeling**. Without coding, every symbol carries $\log_2 M$ bits and

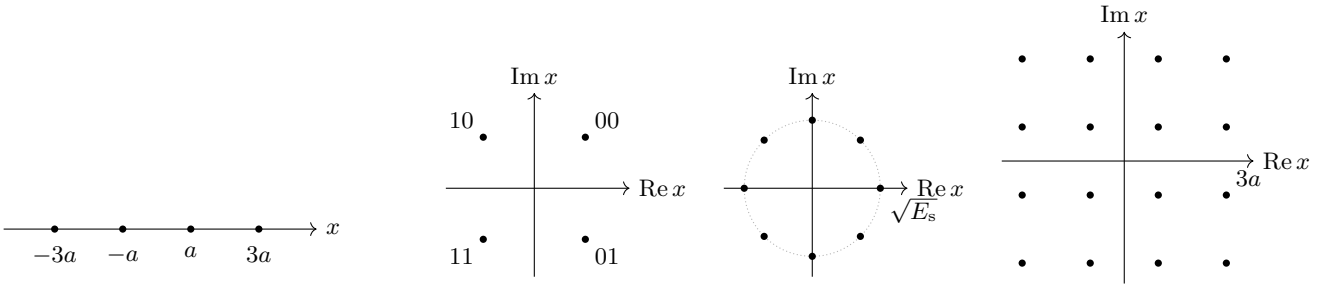
$$R_\mathrm{b} = R_\mathrm{s} \log_2 M. \tag{5.1}$$

With a binary code of rate r before the mapper, the coded bits arrive $\log_2 M$ per symbol but the information bits do not. By equation (3.30) of Topic 3, each symbol carries $R = r \log_2 M$ information bits, so $R_\mathrm{b} = R\, R_\mathrm{s}$. The labeling matters for bit errors, as Section 1.6 of Topic 1 showed: with a Gray labeling, in which neighboring points differ in one bit, the bit error probability is approximately $P_\mathrm{e}/\log_2 M$. The approximation counts only the errors to a neighboring point, which dominate at the SNRs of interest.

The **energy per symbol** is $E_\mathrm{s} = \mathrm{E}(|X|^2)$, as in Topic 1, and the **energy per information bit** is $E_\mathrm{b} = E_\mathrm{s}/R$, which is $E_\mathrm{s}/\log_2 M$ without coding. A symbol of energy E_s every T seconds gives the **average power**

$$P = \frac{E_\mathrm{s}}{T} = E_\mathrm{s} R_\mathrm{s} = E_\mathrm{b} R_\mathrm{b}, \tag{5.2}$$

where the last form, energy per bit times bits per second, does not depend on the constellation or the code. The constellations of this topic are in Figure 5.1 and Table 5.1. The M-PAM constellations are real, with points $\pm a, \pm 3a, \ldots, \pm(M-1)a$ and energy $E_\mathrm{s} = (M^2-1)a^2/3$ by Exercise 1.2 of Topic 1. The M-**PSK** constellations are complex, with M points on a circle of radius $\sqrt{E_\mathrm{s}}$, at angles $2\pi k/M$. The square M-**QAM** constellations, with $M = 4^q$, take the real and imaginary parts from the same $\sqrt{M}$-PAM, so their energy is twice that of the $\sqrt{M}$-PAM: $2a^2$ for QPSK, which is 4-QAM, and $10a^2$ for 16-QAM. The table also gives the squared minimum distance per unit of E_b for uncoded transmission, the quantity that governed the comparisons of Topics 2 and 4. Each additional bit per dimension divides it by a factor of 2.5 to 2.8, about 4 dB. Section 5.6 says what is gained in exchange.



(a) 4-PAM (b) QPSK with Gray labels (c) 8-PSK (d) 16-QAM

Figure 5.1: The constellations of Table 5.1. The spacing between neighbors is $2a$ in (a), (b) and (d).

Constellation	M	$\log_2 M$	Dimensions	E_s	d_min	$d^2_\mathrm{min}/E_\mathrm{b}$
2-PAM	2	1	1	a^2	$2a$	4
4-PAM	4	2	1	$5a^2$	$2a$	1.6
8-PAM	8	3	1	$21a^2$	$2a$	0.57
QPSK	4	2	2	$2a^2$	$2a$	4
8-PSK	8	3	2	E_s	$2\sqrt{E_\mathrm{s}}\sin(\pi/8)$	1.76
16-QAM	16	4	2	$10a^2$	$2a$	1.6
64-QAM	64	6	2	$42a^2$	$2a$	0.57

Table 5.1: Energy and minimum distance of the usual constellations with equiprobable points, and squared minimum distance per unit of energy per bit for uncoded transmission.

Example 5.1. A link carries $R_\mathrm{b} = 12$ Mbit/s with 16-QAM. Without coding, by (5.1) the symbol rate is $R_\mathrm{s} = 12/4 = 3$ Msymbols per second and $T = \frac{1}{3}\,\mu$s. With a code of rate $r = \frac{3}{4}$, each symbol carries $R = \frac{3}{4}\cdot 4 = 3$ information bits, so $R_\mathrm{s} = 4$ Msymbols per second and $T = 0.25\,\mu$s. The code costs one third more symbols per second, which Section 5.6 converts into bandwidth. Coordinates have the units of the square root of an energy, since the basis waveforms have unit energy, so if $a^2 = 1$ nJ, Table 5.1 gives $E_\mathrm{s} = 10$ nJ in both cases, the energy per information bit is $E_\mathrm{b} = 2.5$ nJ without coding and $\frac{10}{3}$ nJ with it, and by (5.2) the power is $P = 30$ mW without coding and 40 mW with it.

5.2 Pulse amplitude modulation

The simplest basis of Topic 4 was a pulse and its shifts. We now take this basis seriously, with pulses that may overlap. Since this topic is full of shifts and reversals of the pulse, we make the time variable explicit in the inner products of Topic 4 and write $\langle \phi(t), \phi(t-\tau)\rangle$ instead of $\langle \phi, \phi(\cdot - \tau)\rangle$: the variable inside the brackets is the one integrated out, and when t is needed for another instant the integration variable is called u.

Definition 5.2. A real signal ϕ of unit energy is a **Nyquist pulse** for the symbol period T when its shifts by multiples of T are orthonormal, that is, when

$$g(\tau) = \langle \phi(t), \phi(t-\tau)\rangle = \int_{-\infty}^{\infty} \phi(t)\,\phi(t-\tau)\,\mathrm{d}t \tag{5.3}$$

satisfies $g(0) = 1$ and $g(kT) = 0$ for every integer $k \neq 0$.

The function g compares the pulse with itself displaced by τ, and the inner product of two shifts, $\langle \phi(t-iT), \phi(t-kT)\rangle$, equals $g((k-i)T)$ because a common shift does not change an inner product, as seen in Section 4.1 of Topic 4. The condition therefore says exactly that $\{\phi(t-iT)\}$ is an orthonormal set. **Pulse amplitude modulation** (PAM) sends the symbols $x_1, \ldots, x_n$ of a real constellation as

$$x(t) = \sum_{i=1}^{n} x_i\,\phi(t-iT), \tag{5.4}$$

one coordinate in each shift of the pulse, at the rate of one symbol every T seconds. Figure 5.2 shows a PAM signal. Everything that Topic 4 proved for an orthonormal basis applies at once. By Proposition 4.4 of Topic 4

the energy of the signal is $\sum_{i=1}^{n} x_i^2$, with average nE_s, and the coordinate x_i is the correlation $\langle x(t), \phi(t-iT)\rangle$. In white Gaussian noise of spectral density $N_0/2$, the correlator outputs are

$$Y_i = \langle Y(t), \phi(t-iT)\rangle = x_i + Z_i, \qquad i = 1, \ldots, n, \tag{5.5}$$

with independent Gaussian noise of variance $N_0/2$ by Proposition 4.10, and nothing is lost by Theorem 4.12. This is the real Gaussian channel of Table 1.2 of Topic 1, with $\sigma^2 = N_0/2$ and $\mathrm{SNR} = 2E_s/N_0$ by equation (4.12) of Topic 4, used $R_s = 1/T$ times per second. The detectors, decoders, error probabilities and mutual informations of Topics 1 to 3 apply without change, and the symbol rate converts their results from channel uses to seconds.

Example 5.3. The rectangular pulse $\phi(t) = \frac{1}{\sqrt{T}}$ for $0 \le t < T$, and 0 otherwise, is a Nyquist pulse for the period T, since its shifts have disjoint supports. Its function g is the triangle of Figure 5.2,

$$g(\tau) = \begin{cases} 1 - \dfrac{|\tau|}{T} & |\tau| \le T \\ 0 & \text{otherwise,} \end{cases} \tag{5.6}$$

because for $0 \le \tau \le T$ the pulse and its shift overlap on an interval of length $T - \tau$, where their product is $\frac{1}{T}$. The triangle vanishes at every nonzero multiple of T, as required, and it also shows what happens between them: the pulse and its shift by half a period have inner product $\frac{1}{2}$.

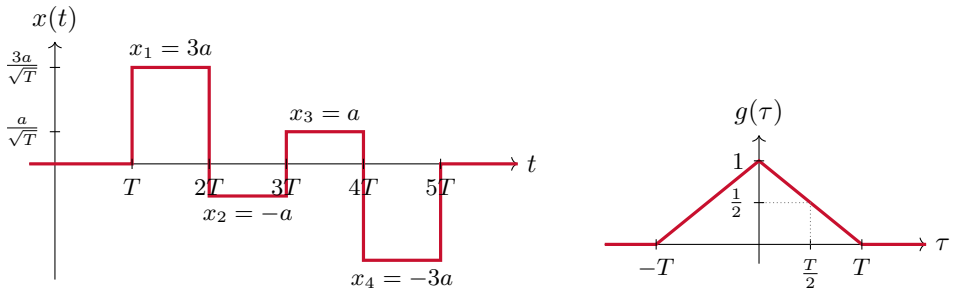


Figure 5.2: The 4-PAM signal (5.4) with the rectangular pulse of Example 5.3: the symbol x_i occupies $[iT, (i+1)T)$. On the right, the function g of that pulse.

Example 5.4. The pulse

$$\phi(t) = \frac{1}{\sqrt{T}} \operatorname{sinc}\Big(\frac{t}{T}\Big), \qquad \operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \tag{5.7}$$

with $\operatorname{sinc}(0) = 1$, is also a Nyquist pulse for the period T. Its shifts overlap over the whole real line, as Figure 5.3 shows, and yet they are orthonormal, like ϕ_1 and ϕ_2 of Example 4.3 of Topic 4. Section 5.5 proves it and shows that $g(\tau) = \operatorname{sinc}(\tau/T)$, which vanishes at every nonzero multiple of T. The pulse is not zero outside any finite interval, and the figure shows the reason for using it: the PAM signal passes through the value $x_i/\sqrt{T}$ at $t = iT$, because all the other pulses vanish there, and it occupies a narrow band of frequencies. As the remark that closed Topic 4 says, the two pulses give the same coordinates, noise and error probabilities, and differ only in the resources they use: the first is confined in time and spread in frequency, the second the opposite.

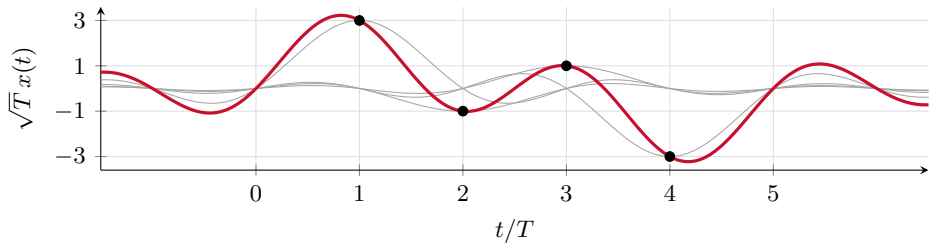


Figure 5.3: The 4-PAM signal of Figure 5.2, now with the sinc pulse of Example 5.4, in units of a. The four scaled pulses (thin) overlap everywhere, and their sum (thick) equals each symbol at its own instant iT.

5.3 Matched filter and sampling

The receiver of Figure 4.4 of Topic 4 has one correlator per basis waveform, that is, one per symbol. For a PAM signal the basis waveforms are shifts of one pulse, and one filter computes all the correlations. The **convolution** of

two signals is

$$(y * h)(t) = \int_{-\infty}^{\infty} y(u)\, h(t-u)\, \mathrm{d}u, \tag{5.8}$$

and a linear time-invariant filter with impulse response h produces $y * h$ from its input y, as seen in the signals and systems course.

Proposition 5.5. Let $h(t) = \phi(-t)$ be the pulse reversed in time. For every signal y and every instant t,

$$(y * h)(t) = \langle y(u), \phi(u-t) \rangle, \tag{5.9}$$

so the output of the filter h at the instant iT is the correlation $\langle y(t), \phi(t-iT) \rangle$ with the i-th basis waveform.

Proof. In (5.8), $h(t-u) = \phi(u-t)$, so $(y*h)(t) = \int_{-\infty}^{\infty} y(u)\phi(u-t)\,\mathrm{d}u$, which is the inner product of y with the pulse shifted by t. □

The filter h is the **matched filter** of the pulse, and Figure 5.4 shows the receiver: the received waveform enters the matched filter, a sampler reads the output at $t = T, 2T, \ldots, nT$, and the samples $Y_1, \ldots, Y_n$ of (5.5) go to the detector or decoder of Topics 1 and 2. The name is justified by the fact that, among all filters followed by a sampler, the matched filter gives the largest ratio of signal to noise in the sample.

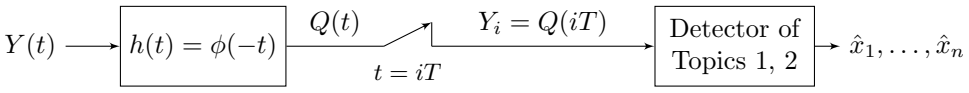


Figure 5.4: The PAM receiver. The matched filter and the sampler compute the n correlations of Figure 4.4 of Topic 4.

The output of the filter between the sampling instants tells what the Nyquist condition protects us from. Without noise, the input is the PAM signal (5.4), and by (5.9), linearity and the definition (5.3),

$$q(t) = \sum_{k=1}^{n} x_k \, \langle \phi(u-kT), \phi(u-t) \rangle = \sum_{k=1}^{n} x_k \, g(t-kT). \tag{5.10}$$

The filter output is a PAM signal with the pulse g in the place of ϕ. At $t = iT$ every term with $k \neq i$ vanishes and $q(iT) = x_i$, which is (5.5) without noise. At an instant $iT + \tau$ with $0 < \tau < T$, the sample is

$$q(iT+\tau) = x_i \, g(\tau) + \sum_{k \neq i} x_k \, g\big((i-k)T + \tau\big), \tag{5.11}$$

a reduced copy of x_i plus a combination of the other symbols, called **intersymbol interference**. It is not noise: it is present without noise, it depends on the data, and it would vanish if the receiver sampled at the right instants. The pulse shape decides how much of it a timing error collects. The sinc pulse has $g(\tau) = \mathrm{sinc}(\tau/T)$, whose values at $\pm T + \tau, \pm 2T + \tau, \ldots$ decrease like $1/k$, and Section 5.5 introduces pulses whose g decays faster, at the price of some bandwidth.

Problem 5.6. The 4-PAM sequence $(x_1, x_2, x_3, x_4) = (3a, -a, a, -3a)$ of Figure 5.2 is sent with the rectangular pulse, and the receiver samples the matched filter at $t = iT + \frac{T}{3}$ instead of $t = iT$. Find the four samples and the decisions of the nearest-point detector, without noise.

Solution: By (5.6), $g(\frac{T}{3}) = \frac{2}{3}$ and $g(-T + \frac{T}{3}) = g(-\frac{2T}{3}) = \frac{1}{3}$, while g vanishes at $T + \frac{T}{3}$ and beyond. In (5.11) only the symbol x_i and the next one, x_{i+1}, survive, so $q(iT + \frac{T}{3}) = \frac{2}{3}x_i + \frac{1}{3}x_{i+1}$, with $x_5 = 0$ after the last symbol. The four samples are $\frac{2}{3}\cdot 3a - \frac{1}{3}a = \frac{5}{3}a$, $-\frac{2}{3}a + \frac{1}{3}a = -\frac{1}{3}a$, $\frac{2}{3}a - a = -\frac{1}{3}a$ and $-2a$. The thresholds of 4-PAM are at 0 and $\pm 2a$, by Problem 1.12 of Topic 1, so the decisions are a, $-a$ and $-a$ for the first three symbols, and the last sample lies exactly on the threshold between $-a$ and $-3a$. A detector must fix a rule for ties, for instance the point of smaller energy, here $-a$, and we adopt it. The first, the third and the fourth symbols are then wrong before any noise is added. The noise in a sample taken at $iT + \tau$ has the same variance $N_0/2$ as at iT, since $\phi(t - iT - \tau)$ has unit energy, so the timing error costs signal and gains nothing.

5.4 Passband transmission: PSK and QAM

A PAM signal built on a slowly varying pulse has its frequency content around $f = 0$, and we call it a **baseband** signal. Radio channels, antennas and the regulations that share the spectrum require the signal to occupy a band around a **carrier frequency** f_c, typically far above $1/T$. Such a signal is a **passband** signal, and the way to build it is to multiply a baseband pulse by a cosine. A sine at the same frequency gives a second pulse, orthogonal to the first, and the two together carry two coordinates per symbol period. Let ψ be a real Nyquist pulse for the period T. The **in-phase** and **quadrature** waveforms of the i-th symbol period are

$$\phi_{\mathrm{I},i}(t) = \sqrt{2}\,\psi(t - iT)\cos(2\pi f_c t) \qquad \text{and} \qquad \phi_{\mathrm{Q},i}(t) = -\sqrt{2}\,\psi(t - iT)\sin(2\pi f_c t). \tag{5.12}$$

The pulse moves from period to period and the carrier does not: its phase is counted from the common origin $t = 0$ in every period, so the whole signal rides on one cosine and one sine, and the receiver will multiply by that same cosine and sine.

Let ψ be the rectangular pulse of Example 5.3 and let f_cT be a positive integer. Then the waveforms $\phi_{\mathrm{I},i}$ and $\phi_{\mathrm{Q},k}$ form an orthonormal set for all integers i and k; the proof, a direct computation with the double-angle formulas, is left to the reader. The same orthonormality holds for any real Nyquist pulse $\psi(t)$ whose baseband bandwidth W satisfies $W < f_c$. In this case, the condition that f_cT be an integer is not needed.

Quadrature amplitude modulation (QAM) sends the complex symbols $x_i = a_i + jb_i$ of a constellation $\mathcal{X} \subset \mathbb{C}$ by placing their real parts on the in-phase waveforms and their imaginary parts on the quadrature waveforms,

$$x(t) = \sum_{i=1}^{n} \big(a_i\,\phi_{\mathrm{I},i}(t) + b_i\,\phi_{\mathrm{Q},i}(t)\big). \tag{5.13}$$

This is a transmitter of Section 4.3 of Topic 4 with a basis of $2n$ real waveforms, and the coordinate vector $(a_1, b_1, \ldots, a_n, b_n)$. The complex number x_i is a convenient name for the pair (a_i, b_i), and no complex waveform is transmitted. Its modulus and argument have a direct meaning. Writing $x_i = |x_i|e^{j\theta_i}$ and using (5.12) with $\cos(u + \theta) = \cos u \cos\theta - \sin u \sin\theta$, the i-th term of (5.13) is

$$a_i\,\phi_{\mathrm{I},i}(t) + b_i\,\phi_{\mathrm{Q},i}(t) = \sqrt{2}\,|x_i|\,\psi(t - iT)\cos\big(2\pi f_c t + \theta_i\big), \tag{5.14}$$

a piece of carrier whose amplitude is set by $|x_i|$ and whose phase is set by θ_i. The symbol modulates the amplitude and the phase of the carrier at once, which gives the method its name. **Phase shift keying** (PSK) is the special case of a constellation on a circle: the amplitude is constant and only the phase carries information. With the rectangular pulse, a QPSK signal is a carrier of constant amplitude $\sqrt{2E_s/T}$ whose phase jumps among $\pm\frac{\pi}{4}$ and $\pm\frac{3\pi}{4}$ at the multiples of T. By Proposition 4.4 of Topic 4, the energy of the signal is $\sum_{i=1}^{n}(a_i^2 + b_i^2) = \sum_{i=1}^{n}|x_i|^2$, so $|x_i|^2$ is the energy of the symbol, in agreement with the definition $E_s = \mathrm{E}(|X|^2)$ of Topic 1 for complex constellations.

Since the waveforms are orthonormal, everything of Topic 4 applies again. The receiver correlates $Y(t)$ with $\phi_{\mathrm{I},i}(t)$ and $\phi_{\mathrm{Q},i}(t)$, which Exercise 5.6 shows is done by multiplying by the carrier, filtering with $\psi(-t)$ and sampling at $t = iT$, and obtains

$$Y_i = x_i + Z_i, \qquad i = 1, \ldots, n, \tag{5.15}$$

as in (5.5), but now with complex numbers: the real and imaginary parts of Z_i are independent Gaussian noises of variance $N_0/2$ each, so the complex noise Z_i has variance N_0. This is the complex Gaussian channel of Table 1.2 of Topic 1 with $\sigma^2 = N_0$ and $\mathrm{SNR} = E_s/N_0$, used once per symbol period.

Example 5.7. With QPSK, $E_s = 2a^2$ and the nearest-point detector of Proposition 1.5 of Topic 1 decides each coordinate from its sign. Each coordinate is 2-PAM with amplitude a in noise of variance $N_0/2$, so by equation (1.23) of Topic 1 it is wrong with probability $p = \frac{1}{2}\operatorname{erfc}\big(a/\sqrt{N_0}\big)$, and by Problem 1.13 and Exercise 1.5 of Topic 1 the symbol error probability is $2p - p^2$ and the bit error probability with Gray labels is exactly p. Since each symbol carries two bits, $E_b = a^2$ and

$$P_b = \tfrac{1}{2}\operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right), \tag{5.16}$$

the same as antipodal signaling in Example 4.14 of Topic 4. The quadrature pulse carries a second bit at no cost in energy per bit, as the four corners of Problem 4.15 of Topic 4 did.

☞ *The passband channel is the complex Gaussian channel, so Theorem 3.14 of Topic 3 applies with* $\mathrm{SNR} = E_s/N_0$*: at most* $\log_2(1 + E_s/N_0)$ *bits per symbol, that is,* $R_s \log_2(1 + E_s/N_0)$ *bits per second, can be sent reliably. Topic 6 replaces* R_s *by the bandwidth.*

5.5 Bandwidth and the Nyquist criterion

Everything so far was measured in time. The second resource of a waveform is the band of frequencies it occupies, and the tool to measure it is the Fourier transform of the signals and systems course.

Definition 5.8. The **Fourier transform** of a signal x is

$$\hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, \mathrm{d}t, \tag{5.17}$$

a complex function of the frequency f, in hertz, and $x(t) = \int_{-\infty}^{\infty} \hat{x}(f) e^{j2\pi ft}\, \mathrm{d}f$ recovers the signal.

Table 5.2 lists the properties we use: shifting, filtering and multiplying by a carrier act simply on $\hat{x}$, and Parseval's identity computes inner products and energies in either domain, so $|\hat{x}(f)|^2$ is the density of the energy of x in frequency. The first pair at the end is computed in Example 5.9; the second is its dual, with time and frequency exchanged.

Property	Signal	Transform
Shift	$x(t-\tau)$	$e^{-j2\pi f\tau}\,\hat{x}(f)$
Convolution	$(x * h)(t)$	$\hat{x}(f)\,\hat{h}(f)$
Modulation	$x(t)\cos(2\pi f_{\mathrm{c}} t)$	$\frac{1}{2}\hat{x}(f-f_{\mathrm{c}}) + \frac{1}{2}\hat{x}(f+f_{\mathrm{c}})$
Parseval	$\langle x(t), y(t)\rangle$	$\int_{-\infty}^{\infty} \hat{x}(f)\,\hat{y}(f)^*\,\mathrm{d}f$
Real signal	$x(t)$ real	$\hat{x}(-f) = \hat{x}(f)^*$
Reversal	$x(-t)$, x real	$\hat{x}(f)^*$
Rectangular pulse	$\frac{1}{\sqrt{T}}$ on $[-\frac{T}{2}, \frac{T}{2})$	$\sqrt{T}\,\mathrm{sinc}(fT)$
Sinc pulse	$\frac{1}{\sqrt{T}}\,\mathrm{sinc}(t/T)$	$\sqrt{T}$ on $[-\frac{1}{2T}, \frac{1}{2T})$

Table 5.2: Properties of the Fourier transform and the two transform pairs of this topic. The asterisk denotes the complex conjugate.

Example 5.9. For the rectangular pulse of unit energy centered at the origin, $x(t) = \frac{1}{\sqrt{T}}$ for $-\frac{T}{2} \le t < \frac{T}{2}$, the integral (5.17) is, for $f \neq 0$,

$$\hat{x}(f) = \frac{1}{\sqrt{T}} \int_{-T/2}^{T/2} e^{-j2\pi ft}\, \mathrm{d}t = \frac{1}{\sqrt{T}}\, \frac{e^{-j\pi fT} - e^{j\pi fT}}{-j2\pi f} = \frac{1}{\sqrt{T}}\, \frac{\sin(\pi fT)}{\pi f} = \sqrt{T}\,\mathrm{sinc}(fT), \tag{5.18}$$

and $\hat{x}(0) = \sqrt{T}$ is the integral of the pulse, which agrees with $\mathrm{sinc}(0) = 1$. The energy density $|\hat{x}(f)|^2 = T\,\mathrm{sinc}^2(fT)$ decays like $1/f^2$. The spectrum extends to arbitrarily high frequencies, despite its zeros at nonzero integer multiples of $1/T$, with about 90% of the energy in $|f| < 1/T$ and 95% in $|f| < 2/T$. The pulse of Example 5.3 is this one shifted by $\frac{T}{2}$. A time shift multiplies the Fourier transform by a phase factor, leaving its magnitude and energy density unchanged.

Definition 5.10. A baseband signal x has **bandwidth** W when $\hat{x}(f) = 0$ for $|f| > W$ and W is the smallest such number. A passband signal whose transform vanishes outside $f_{\mathrm{c}} - W \le |f| \le f_{\mathrm{c}} + W$ has **passband bandwidth** $2W$.

For a real signal the negative frequencies repeat the positive ones, by the fifth line of Table 5.2, so the bandwidth is the extent of the positive frequencies. By the modulation property, the passband pulses (5.12) built on a pulse ψ of bandwidth W occupy $f_{\mathrm{c}} - W \le |f| \le f_{\mathrm{c}} + W$, a passband bandwidth of $2W$, and by linearity and the shift property every PAM or QAM signal has the bandwidth of its pulse. The sinc pulse has bandwidth $1/(2T)$ and the rectangular pulse has no finite bandwidth; the theorem below says how small the bandwidth of a Nyquist pulse can be. It rests on one identity: by Parseval's identity and the shift property, the function (5.3) is

$$g(\tau) = \int_{-\infty}^{\infty} \hat{\phi}(f)\, \big(e^{-j2\pi f\tau}\hat{\phi}(f)\big)^*\, \mathrm{d}f = \int_{-\infty}^{\infty} |\hat{\phi}(f)|^2\, e^{j2\pi f\tau}\, \mathrm{d}f, \tag{5.19}$$

the inverse Fourier transform of the energy density of the pulse.

Theorem 5.11 (Nyquist criterion)**.** A real pulse ϕ of unit energy is a Nyquist pulse for the period T if and only if

$$\sum_{k=-\infty}^{\infty} \Big|\hat{\phi}\Big(f - \frac{k}{T}\Big)\Big|^2 = T \qquad \text{for almost every } f. \tag{5.20}$$

Consequently, a Nyquist pulse for the period T has bandwidth at least $1/(2T)$, and a Nyquist pulse with bandwidth exactly $1/(2T)$ has the energy density of the sinc pulse (5.7), $|\hat{\phi}(f)|^2 = T$ for almost every f with $|f| < 1/(2T)$.

Values at isolated spectral endpoints do not affect the pulse energy or the orthogonality of its shifts. The theorem follows from (5.19): the samples $g(kT)$ are the Fourier series coefficients of the periodic function in (5.20), and all but $g(0)$ vanish exactly when it is constant. For the sinc pulse, $|\hat{\phi}(f)|^2$ is the rectangle of height T on $|f| < 1/(2T)$, whose shifted copies tile the frequency axis, and g, its inverse transform by (5.19), is $\operatorname{sinc}(\tau/T)$, as Example 5.4 claimed. Conversely, the rectangular pulse is a Nyquist pulse by Example 5.3, so the shifted copies of $\operatorname{sinc}^2(fT)$ add up to 1 at every frequency, which is not obvious from the formula.

The theorem is a statement about resources. A baseband waveform of bandwidth W can carry a Nyquist pulse of period T only if $W \geq 1/(2T)$, that is,

$$R_{\mathrm{s}} = \frac{1}{T} \leq 2W \tag{5.21}$$

symbols per second, with equality for the sinc pulse. In passband, where the bandwidth is $2W$ by Definition 5.10, the same pulse carries $R_{\mathrm{s}} \leq 2W$ complex symbols per second, with two real coordinates each. In both cases a hertz of bandwidth buys two real dimensions per second, the count that Topic 6 uses. The sinc pulse attains the limit, but its slow decay makes it fragile to timing errors, and practical systems spend some **excess bandwidth** on pulses that decay faster.

Example 5.12. The **raised-cosine** family with **roll-off** $\beta \in [0, 1]$ is defined by the energy density

$$|\hat{\phi}(f)|^2 = \begin{cases} T & |f| \leq \dfrac{1-\beta}{2T} \\ \dfrac{T}{2}\Big(1 + \cos\Big(\dfrac{\pi T}{\beta}\Big(|f| - \dfrac{1-\beta}{2T}\Big)\Big)\Big) & \dfrac{1-\beta}{2T} < |f| < \dfrac{1+\beta}{2T} \\ 0 & \text{otherwise,} \end{cases} \tag{5.22}$$

shown in Figure 5.5. The bandwidth is $W = (1+\beta)/(2T)$, so the excess over the minimum $1/(2T)$ is the fraction β. For $\beta = 0$ this is the sinc pulse, and for $\beta = 1$ the bandwidth is doubled. The pulse satisfies (5.20) because the copies $k = 0$ and $k = 1$ overlap on the transition band around $f = 1/(2T)$ with opposite cosines, as Exercise 5.5 checks for $\beta = 1$. The function g is the inverse transform of (5.22),

$$g(\tau) = \operatorname{sinc}\Big(\frac{\tau}{T}\Big)\,\frac{\cos(\pi\beta\tau/T)}{1 - (2\beta\tau/T)^2}, \tag{5.23}$$

which decays like $1/\tau^3$ for $\beta > 0$, and the pulse ϕ itself, whose transform is the square root of (5.22), is called the root raised-cosine pulse. With $\beta = 1$ and the timing error of Problem 5.6, the interference coefficients $g(-\frac{2T}{3}) = 0.27$ and $g(\frac{4T}{3}) = -0.02$ replace the 0.41 and -0.21 of the sinc pulse.

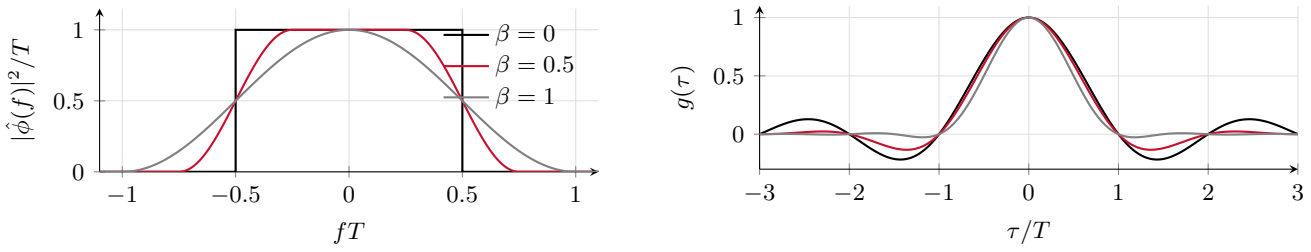


Figure 5.5: The raised-cosine family of Example 5.12 for $\beta = 0$, 0.5 and 1: energy density in frequency (left) and the function g (right). All three vanish at the nonzero multiples of T, and the larger β, the faster g decays and the wider the band.

5.6 Spectral efficiency and the three resources

We can now count what a transmission costs in bandwidth. The **spectral efficiency** of a system is

$$\eta = \frac{R_{\mathrm{b}}}{W_{\mathrm{total}}} \tag{5.24}$$

bits per second per hertz, where W_{total} is the bandwidth it occupies, baseband or passband. With a raised-cosine pulse of roll-off β and R information bits per symbol, baseband PAM has $W_{\text{total}} = (1+\beta)R_{\text{s}}/2$ and passband QAM has $W_{\text{total}} = (1+\beta)R_{\text{s}}$, so

$$\eta_{\text{PAM}} = \frac{2R}{1+\beta} \qquad \text{and} \qquad \eta_{\text{QAM}} = \frac{R}{1+\beta}, \tag{5.25}$$

with $R = \log_2 M$ without coding. The factor 2 is not an advantage of baseband: both systems use the same $2W_{\text{total}}/(1+\beta)$ real dimensions per second, out of the limit $2W_{\text{total}}$ of (5.21), and 16-QAM against 4-PAM, at equal bandwidth, have the same η and the same $d^2_{\text{min}}/E_{\text{b}}$ in Table 5.1.

The trade-off concerns energy per bit E_{b} (power $P = E_{\text{b}}R_{\text{b}}$), bandwidth (through η), and reliability (P_{b}). Table 5.3 places the constellations of this topic in this trade-off for uncoded transmission with Gray labels, with P_{b} from Problem 1.12 and Exercise 1.2 of Topic 1 applied to each real coordinate, as in Example 5.7. Each step from M to $4M$ points in QAM, one more bit per real dimension, adds $2/(1+\beta)$ bits per second per hertz, 2 for the sinc pulse of the table, and costs about 4 dB of energy per bit at the same bit error probability: energy and bandwidth are traded against each other, and the bit error probability sets the exchange rate. Coding enters the same trade-off: $R = r\log_2 M$ reduces η by the factor r, and a good code reduces the E_{b}/N_0 required for the same P_{b}, by at most what Theorem 3.14 of Topic 3 allows, which Topic 6 writes in these units. Exercise 5.7 works one case.

Constellation	η with $\beta = 0$	P_{b} with Gray labels	E_{b}/N_0 for $P_{\text{b}} = 10^{-3}$	E_{b}/N_0 for $P_{\text{b}} = 10^{-5}$
QPSK	2	$\frac{1}{2}\operatorname{erfc}\left(\sqrt{E_{\text{b}}/N_0}\right)$	6.8 dB	9.6 dB
16-QAM	4	$\frac{3}{8}\operatorname{erfc}\left(\sqrt{2E_{\text{b}}/(5N_0)}\right)$	10.5 dB	13.4 dB
64-QAM	6	$\frac{7}{24}\operatorname{erfc}\left(\sqrt{E_{\text{b}}/(7N_0)}\right)$	14.8 dB	17.8 dB

Table 5.3: Spectral efficiency of uncoded passband QAM with the sinc pulse, bit error probability with Gray labels and nearest-point detection, where only errors to a neighboring point are counted for 16-QAM and 64-QAM, and the energy per bit needed for two targets. Baseband 2-PAM, 4-PAM and 8-PAM have the same three last columns.

Wireless local area networks and digital television adapt the constellation to the link SNR (Table 5.3). They also transmit QAM symbols on parallel, orthogonal **subcarriers** spaced $1/T$ apart and lasting T seconds, despite overlapping spectra. This is **orthogonal frequency division multiplexing** (OFDM), Topic 4 applied in frequency.

☞ *The pulse and the carrier decide where in time and frequency the coordinates travel. The constellation and the code decide how many bits each coordinate carries, and how reliably. Topic 4 proved that the second question is independent of the first. This topic has priced the first, two real dimensions per hertz per second, and Topic 6 asks how many bits those dimensions can carry.*

What to remember

- A symbol every T seconds carries $R = r\log_2 M$ information bits, so $R_{\text{b}} = R\,R_{\text{s}}$, $E_{\text{b}} = E_{\text{s}}/R$ and $P = E_{\text{s}}R_{\text{s}} = E_{\text{b}}R_{\text{b}}$.
- PAM sends $x(t) = \sum_i x_i\phi(t-iT)$ with a Nyquist pulse, one whose shifts by T are orthonormal, that is, $g(0) = 1$ and $g(kT) = 0$ for $k \neq 0$. The receiver is the matched filter $\phi(-t)$ sampled at iT, and it sees the real Gaussian channel with $\sigma^2 = N_0/2$, R_{s} times per second. Sampling at the wrong instants adds intersymbol interference, $\sum_{k\neq i} x_k g((i-k)T+\tau)$.
- QAM places $\operatorname{Re} x_i$ on $\sqrt{2}\psi(t-iT)\cos(2\pi f_{\text{c}}t)$ and $\operatorname{Im} x_i$ on $-\sqrt{2}\psi(t-iT)\sin(2\pi f_{\text{c}}t)$. The symbol sets the amplitude and the phase of the carrier, PSK uses the phase alone, and the receiver sees the complex Gaussian channel with $\sigma^2 = N_0$ and $\text{SNR} = E_{\text{s}}/N_0$.
- The bandwidth of a baseband signal is the extent of its positive frequencies, and a passband signal built on a pulse of bandwidth W occupies $2W$ around f_{c}. The Nyquist criterion, $\sum_k |\hat{\phi}(f-k/T)|^2 = T$, forces $W \geq 1/(2T)$, so $R_{\text{s}} \leq 2W$ symbols per second: a hertz buys two real dimensions per second. The sinc pulse attains the limit and the raised-cosine family trades excess bandwidth β for a faster decay in time.
- With $\eta = R_{\text{b}}/W_{\text{total}}$, baseband PAM gives $\eta = 2R/(1+\beta)$ and passband QAM gives $\eta = R/(1+\beta)$. Changing QAM from M to $4M$ adds $2/(1+\beta)$ bits/s/Hz, costing about 4 dB of E_{b}/N_0 at fixed P_{b}.

Exercises

Exercise 5.1. A satellite link sends $R_\mathrm{b} = 6$ Mbit/s with 8-PSK and a code of rate $r = \frac{2}{3}$. Find the information bits per symbol, the symbol rate, the symbol period, and, with $E_\mathrm{s} = 3$ pJ, the energy per information bit and the transmitted power. What changes without the code?

Exercise 5.2. The symbols $(x_1, x_2, x_3) = (a, a, -a)$ are sent with the pulse $\phi(t) = \sqrt{2/T}$ for $0 \le t < \frac{T}{2}$, and 0 otherwise, and the period T. Check that ϕ is a Nyquist pulse for T, draw the PAM signal, compute its energy from the waveform and from the coordinates, and find the function g of this pulse. What does the matched filter output look like at the instants $iT + \frac{T}{2}$?

Exercise 5.3. The 4-PAM sequence $(x_1, x_2, x_3, x_4) = (a, 3a, -3a, -a)$ is sent with the rectangular pulse of Example 5.3, and the receiver samples the matched filter too early, at $t = iT - \frac{T}{4}$. Find the four noise-free samples and the decisions of the nearest-point detector. If the sinc pulse of Example 5.4 is used instead, with the same timing error, which symbols contribute to the i-th sample, and with which coefficients for x_{i-1}, x_i and x_{i+1}?

Exercise 5.4. A baseband link sends 4-PAM at $R_\mathrm{s} = 10$ Msymbols per second with root raised-cosine pulses of roll-off $\beta = 0.5$. Find its bandwidth, bit rate and spectral efficiency. The same pulses and symbol rate are then used in passband with 16-QAM on the carrier $f_\mathrm{c} = 2.4$ GHz: give the band occupied, the bit rate and the spectral efficiency. Finally, find the largest roll-off that lets uncoded 16-QAM send 50 Mbit/s in a passband channel of 15 MHz.

Exercise 5.5. For $\beta = 1$, write (5.22) explicitly, check the Nyquist criterion (5.20) by adding the copies $k = 0$ and $k = 1$ for $0 \le f \le \frac{1}{T}$, and give the bandwidth, the largest symbol rate in a baseband channel of 10 MHz and the spectral efficiency of 4-PAM on that channel. Check that (5.23) with $\beta = 1$ vanishes at every nonzero multiple of T.

Exercise 5.6. Show that multiplying the received signal by $\sqrt{2}\cos(2\pi f_\mathrm{c} t)$ and filtering the product with $\psi(-t)$, sampled at $t = iT$, gives $\langle Y(t), \phi_{\mathrm{I},i}(t)\rangle$. Explain which multiplier gives the quadrature sample. Then let the receiver use the carrier phase θ by mistake, multiplying by $\sqrt{2}\cos(2\pi f_\mathrm{c} t + \theta)$ and $-\sqrt{2}\sin(2\pi f_\mathrm{c} t + \theta)$. Show that, for the rectangular pulse with $f_\mathrm{c} T$ integer, the noise-free samples are $x_i e^{-j\theta}$, and describe the effect on QPSK for $\theta = \frac{\pi}{4}$.

Exercise 5.7. A passband channel of 10 MHz must carry 30 Mbit/s with root raised-cosine pulses of roll-off $\beta = 0.25$ and a bit error probability of at most 10^{-5}. Find the symbol rate, the number of information bits per symbol, and the smallest constellation of Table 5.3 that suffices without coding, with its required E_b/N_0 and its power for $N_0 = 4 \cdot 10^{-21}$ J. Then propose a combination of a larger constellation and a code of rate r that reaches the same bit rate, and say what is known and what is not about its required E_b/N_0.

Solutions

Exercise 5.1. By equation (3.30) of Topic 3, each 8-PSK symbol carries $R = r \log_2 M = \frac{2}{3} \cdot 3 = 2$ information bits. Since $R_\mathrm{b} = R\, R_\mathrm{s}$,

$$R_\mathrm{s} = \frac{R_\mathrm{b}}{R} = \frac{6 \cdot 10^6}{2} = 3 \text{ Msymbols per second} \qquad \text{and} \qquad T = \frac{1}{R_\mathrm{s}} = \tfrac{1}{3}\,\mu\text{s}. \tag{5.26}$$

The energy per information bit and, by (5.2), the power are

$$E_\mathrm{b} = \frac{E_\mathrm{s}}{R} = 1.5 \text{ pJ} \qquad \text{and} \qquad P = E_\mathrm{s} R_\mathrm{s} = 3 \cdot 10^{-12} \cdot 3 \cdot 10^6 = 9\,\mu\text{W} = E_\mathrm{b} R_\mathrm{b}. \tag{5.27}$$

Without the code, $R = 3$, $R_\mathrm{s} = 2$ Msymbols per second, $T = 0.5\,\mu$s, $E_\mathrm{b} = 1$ pJ and $P = 6\,\mu$W. The code spends half as many symbols again per second and one and a half times the energy per information bit. What it buys in reliability is the subject of Topics 2 and 3, and what the extra symbols cost in bandwidth is (5.25).

Exercise 5.2. The pulse has energy $\frac{2}{T} \cdot \frac{T}{2} = 1$, and its shifts by multiples of T occupy the disjoint intervals $[iT, iT + \frac{T}{2})$, so they are orthonormal and ϕ is a Nyquist pulse for the period T. The PAM signal (5.4) equals $a\sqrt{2/T}$ on $[T, \frac{3T}{2})$ and on $[2T, \frac{5T}{2})$, equals $-a\sqrt{2/T}$ on $[3T, \frac{7T}{2})$, and is zero elsewhere, in particular on the

second half of every period, as Figure 5.6 shows. Its energy from the waveform and from the coordinates is

$$\int_{-\infty}^{\infty} x(t)^2 \,\mathrm{d}t = 3 \cdot \frac{2a^2}{T} \cdot \frac{T}{2} = 3a^2 = a^2 + a^2 + a^2, \tag{5.28}$$

in agreement with Proposition 4.4 of Topic 4. The function g is a triangle of base $[-\frac{T}{2}, \frac{T}{2}]$ and height 1, by the computation of Example 5.3 with $\frac{T}{2}$ in the place of T,

$$g(\tau) = \begin{cases} 1 - \dfrac{2|\tau|}{T} & |\tau| \le \frac{T}{2} \\ 0 & \text{otherwise.} \end{cases} \tag{5.29}$$

By (5.10), the matched filter output is $\sum_k x_k g(t - kT)$, a train of triangles of height x_k centered at kT that do not overlap, so at $t = iT + \frac{T}{2}$ the output is exactly zero for every i. This pulse wastes half of each period: with the same bandwidth of its rectangular shape, a pulse of duration $\frac{T}{2}$ could carry twice the symbol rate.

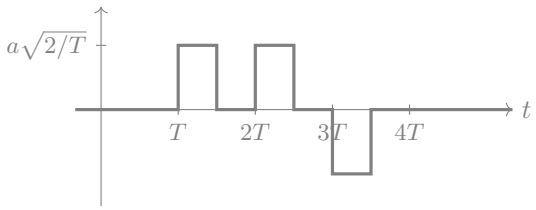


Figure 5.6: The PAM signal of Exercise 5.2: a half-period pulse in each slot, with the sign of the symbol.

Exercise 5.3. By (5.11) with $\tau = -\frac{T}{4}$ and the triangle (5.6), the coefficient of x_i is $g(-\frac{T}{4}) = \frac{3}{4}$, the coefficient of x_{i-1} is $g(T - \frac{T}{4}) = g(\frac{3T}{4}) = \frac{1}{4}$, and g vanishes at $-T - \frac{T}{4}$ and beyond. Sampling early collects the previous symbol instead of the next one,

$$q\big(iT - \tfrac{T}{4}\big) = \tfrac{3}{4}x_i + \tfrac{1}{4}x_{i-1}, \qquad x_0 = 0, \tag{5.30}$$

so the four samples are

$$\tfrac{3}{4}a, \qquad \tfrac{9}{4}a + \tfrac{1}{4}a = \tfrac{5}{2}a, \qquad -\tfrac{9}{4}a + \tfrac{3}{4}a = -\tfrac{3}{2}a, \qquad -\tfrac{3}{4}a - \tfrac{3}{4}a = -\tfrac{3}{2}a. \tag{5.31}$$

With the thresholds 0 and $\pm 2a$ of 4-PAM, the decisions are a, $3a$, $-a$ and $-a$. The third symbol is wrong, pulled toward the origin by the large positive symbol before it, and the other three stay inside their decision regions. With the sinc pulse, $g(\tau) = \operatorname{sinc}(\tau/T)$ vanishes at no instant of the form $kT - \frac{T}{4}$, so every symbol of the sequence contributes to every sample. The coefficients are

$$g\big(-\tfrac{T}{4}\big) = \operatorname{sinc}\big(\tfrac{1}{4}\big) = 0.90 \text{ for } x_i, \quad g\big(\tfrac{3T}{4}\big) = \operatorname{sinc}\big(\tfrac{3}{4}\big) = 0.30 \text{ for } x_{i-1}, \quad g\big(-\tfrac{5T}{4}\big) = \operatorname{sinc}\big(\tfrac{5}{4}\big) = -0.18 \text{ for } x_{i+1}, \tag{5.32}$$

followed by -0.13 for x_{i-2} and 0.10 for x_{i+2}. The interference is spread over many symbols and decays slowly, as the remark after Problem 5.6 announced.

Exercise 5.4. By Example 5.12, the bandwidth of the pulse and the spectral efficiency of uncoded 4-PAM, which carries 2 bits per symbol, are

$$W = \frac{1+\beta}{2T} = \frac{1.5 \cdot 10}{2} = 7.5 \text{ MHz}, \qquad R_\mathrm{b} = 20 \text{ Mbit/s}, \qquad \eta = \frac{20}{7.5} = 2.67 \text{ bits per second per hertz}, \tag{5.33}$$

which is $2R/(1+\beta)$ in (5.25). In passband the signal occupies $f_\mathrm{c} - W \le |f| \le f_\mathrm{c} + W$, from 2392.5 to 2407.5 MHz, a passband bandwidth of 15 MHz. Each 16-QAM symbol carries 4 bits, so

$$R_\mathrm{b} = 40 \text{ Mbit/s} \qquad \text{and} \qquad \eta = \frac{40}{15} = 2.67 \text{ bits per second per hertz} \tag{5.34}$$

again, now $R/(1+\beta)$ with $R = 4$: the passband system sends twice the bits in twice the bandwidth, two real coordinates per symbol instead of one. For 50 Mbit/s with 16-QAM, $R_\mathrm{s} = 12.5$ Msymbols per second, and the

passband bandwidth must not exceed the channel,

$$(1+\beta)R_{\mathrm{s}} \le 15 \text{ MHz} \qquad \Longrightarrow \qquad \beta \le \frac{15}{12.5} - 1 = 0.2. \tag{5.35}$$

A smaller roll-off fits more symbols in the band at the price of a slower decay of g, as Figure 5.5 shows.

Exercise 5.5. With $\beta = 1$ the flat part of (5.22) disappears and

$$|\hat{\phi}(f)|^2 = \frac{T}{2}\big(1+\cos(\pi f T)\big) \quad \text{for } |f| < \tfrac{1}{T}, \qquad 0 \text{ elsewhere.} \tag{5.36}$$

For $0 \le f \le \frac{1}{T}$, only the copies $k = 0$ and $k = 1$ reach the interval, and since $\cos(u-\pi) = -\cos u$,

$$\frac{T}{2}\big(1+\cos(\pi f T)\big) + \frac{T}{2}\Big(1+\cos\big(\pi(f-\tfrac{1}{T})T\big)\Big) = \frac{T}{2}\big(1+\cos(\pi f T)\big) + \frac{T}{2}\big(1-\cos(\pi f T)\big) = T. \tag{5.37}$$

The same holds on every interval $[\frac{k}{T}, \frac{k+1}{T}]$ by periodicity, so (5.20) holds. The bandwidth is $W = \frac{1+\beta}{2T} = \frac{1}{T}$, twice the minimum. In a baseband channel of 10 MHz, $\frac{1}{T} \le 10$ MHz, so the largest symbol rate is $R_{\mathrm{s}} = 10$ Msymbols per second, half of the 20 Msymbols per second that the sinc pulse would allow by (5.21). With 4-PAM, $R_{\mathrm{b}} = 20$ Mbit/s and $\eta = 2$ bits per second per hertz, which is (5.25) with $R = 2$ and $\beta = 1$. In (5.23) with $\beta = 1$,

$$g(\tau) = \operatorname{sinc}\Big(\frac{\tau}{T}\Big)\,\frac{\cos(\pi\tau/T)}{1-4\tau^2/T^2}, \tag{5.38}$$

the first factor vanishes at every nonzero multiple of T, and the second factor is finite there, since its denominator vanishes only at $\tau = \pm\frac{T}{2}$, where the numerator vanishes too and the quotient has the finite limit $\frac{\pi}{4}$. Hence $g(kT) = 0$ for $k \neq 0$, and $g(0) = 1$.

Exercise 5.6. By Proposition 5.5 with the pulse ψ, the filter $\psi(-t)$ sampled at iT computes $\langle v(t), \psi(t-iT)\rangle$ for its input v. With $v(t) = \sqrt{2}\,Y(t)\cos(2\pi f_{\mathrm{c}} t)$, by (5.12) this is

$$\langle v(t), \psi(t-iT)\rangle = \int_{-\infty}^{\infty} Y(t)\,\sqrt{2}\cos(2\pi f_{\mathrm{c}} t)\,\psi(t-iT)\,\mathrm{d}t = \langle Y(t), \phi_{\mathrm{I},i}(t)\rangle, \tag{5.39}$$

the in-phase sample. The multiplier $-\sqrt{2}\sin(2\pi f_{\mathrm{c}} t)$ gives the quadrature sample in the same way. The multiplication moves the signal from the carrier back to baseband, and one filter matched to the baseband pulse does the rest. With the wrong phase, the receiver correlates Y with $\sqrt{2}\psi(t-iT)\cos(2\pi f_{\mathrm{c}} t + \theta)$ and $-\sqrt{2}\psi(t-iT)\sin(2\pi f_{\mathrm{c}} t+\theta)$. Without noise, by (5.14) the i-th term of the signal is $\sqrt{2}|x_i|\psi(t-iT)\cos(2\pi f_{\mathrm{c}} t+\theta_i)$, and the in-phase sample is

$$\frac{2}{T}\int_{iT}^{(i+1)T} |x_i|\cos(2\pi f_{\mathrm{c}} t + \theta_i)\cos(2\pi f_{\mathrm{c}} t + \theta)\,\mathrm{d}t = |x_i|\cos(\theta_i - \theta), \tag{5.40}$$

by $2\cos u\cos v = \cos(u-v) + \cos(u+v)$, since the term at frequency $2f_{\mathrm{c}}$ completes $2f_{\mathrm{c}}T$ whole cycles in the period and integrates to zero. The quadrature sample is $|x_i|\sin(\theta_i - \theta)$ by the same computation with $2\sin u\cos v = \sin(u-v) + \sin(u+v)$, and the sign convention of $\phi_{\mathrm{Q},i}$ gives the sign shown. The complex sample is

$$|x_i|\cos(\theta_i - \theta) + j|x_i|\sin(\theta_i - \theta) = |x_i|e^{j(\theta_i - \theta)} = x_i e^{-j\theta}\,: \tag{5.41}$$

the whole constellation is rotated by $-\theta$. The other symbols contribute nothing, because their pulses occupy other periods. For QPSK and $\theta = \frac{\pi}{4}$, the four points $a(\pm 1 \pm j)$ rotate onto the axes, at $\pm a\sqrt{2}$ and $\pm ja\sqrt{2}$, exactly on the boundaries between two quadrants, at the same distance from two constellation points. With a vanishing amount of noise each of the two is chosen half of the time, so half of the symbols are wrong, and without noise the outcome depends only on the tie rule of the detector. Receivers therefore estimate the carrier phase before deciding, a task called synchronization that is outside this course.

Exercise 5.7. With $\beta = 0.25$, the passband bandwidth of QAM is $1.25R_{\mathrm{s}}$ by (5.25), so the symbol rate and the information bits per symbol needed for 30 Mbit/s are

$$R_{\mathrm{s}} \le \frac{10}{1.25} = 8 \text{ Msymbols per second} \qquad \text{and} \qquad R \ge \frac{30}{8} = 3.75. \tag{5.42}$$

Without coding, $R = \log_2 M$ must be an integer of at least 3.75, so the smallest constellation of Table 5.3 is 16-QAM, with $R_\mathrm{b} = 32$ Mbit/s at $R_\mathrm{s} = 8$ Msymbols per second, or exactly 30 Mbit/s at 7.5 Msymbols per second, and $E_\mathrm{b}/N_0 = 13.4$ dB, that is, $E_\mathrm{b}/N_0 = 22$, for $P_\mathrm{b} = 10^{-5}$. With $N_0 = 4 \cdot 10^{-21}$ J, by (5.2) the power at 30 Mbit/s is

$$E_\mathrm{b} = 22 \cdot 4 \cdot 10^{-21} = 8.8 \cdot 10^{-20} \text{ J} \qquad \text{and} \qquad P = E_\mathrm{b} R_\mathrm{b} = 8.8 \cdot 10^{-20} \cdot 3 \cdot 10^7 = 2.6 \cdot 10^{-12} \text{ W}, \tag{5.43}$$

the power at the receiver, since N_0 describes the noise seen there. An alternative is 64-QAM with a code of rate $r = \frac{5}{8}$, which gives $R = \frac{5}{8} \cdot 6 = 3.75$ and exactly 30 Mbit/s at 8 Msymbols per second. Its required E_b/N_0 is not in the table. Uncoded 64-QAM needs 17.8 dB, so the code must gain more than 4.4 dB to beat 16-QAM. The capacity $\log_2(1 + E_\mathrm{s}/N_0)$ of Theorem 3.14 of Topic 3 does not settle the question, because it is attained with Gaussian inputs and not with a fixed constellation. The right threshold is the mutual information of equiprobable 64-QAM, which Topic 3 identifies as the limit of the rates of codes whose symbols follow that distribution. At the same $E_\mathrm{b}/N_0 = 22$ required by uncoded 16-QAM, the proposed coded 64-QAM system has $E_\mathrm{s}/N_0 = RE_\mathrm{b}/N_0 = 3.75 \cdot 22 = 82.5$. Its mutual information, computed as the curves of Table 3.5 of Topic 3, is approximately 5.7 bits per symbol, above the required 3.75. It reaches 3.75 bits per symbol at $E_\mathrm{s}/N_0 \approx 11.7$ dB, that is, $E_\mathrm{b}/N_0 \approx 6$ dB. Codes of rate $\frac{5}{8}$ over 64-QAM with arbitrarily small error probability therefore exist at any E_b/N_0 above about 6 dB, some 7 dB below uncoded 16-QAM. How close a given code comes to this threshold is a question for a course on coding.